

Paper 1 Progress Update

BLG vs β -Casein Comparative Study at the Air–Water Interface

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Executive Summary

- **BLG dataset is complete and internally consistent** across 4 independent replicas (4.00 μ s total unbiased MD). All four show a stable, compact fold (radius of gyration 1.49–1.50 nm) with no evidence of structural commitment upon contact with the interface.
- **CASEIN now has its second independent replicate (R1) for the first time**, alongside the original CENTER run – all 7 planned analyses complete on both. The N-terminal region’s solvent accessibility reproduces tightly across the two replicas (24.8–25.4 nm²), strengthening the “open chain” comparative claim.
- **The comparative headline holds up with real replication now:** BLG’s calyx stays small and selectively accessible (3.15–3.81 nm²), while CASEIN’s N-terminus is both far larger and almost permanently exposed (contact in 99.5% of frames, in both CENTER and R1) – consistent with “An Accessible Calyx and an Open Chain.”
- **Surface tension cross-validates across species and replicas** (BLG $\approx$ 52 mN/m, CAS $\approx$ 52 mN/m), matching the TIP3P water model’s literature value ($\approx$ 50–52 mN/m) – a good internal consistency check on both force fields.
- **One data-completeness issue found and being corrected:** three of BLG’s four replicas were run in multiple simulation segments; for two analyses (R_g and surface tension), the current scripts only read the first segment (500 of 1000 ns) for those three replicas. RMSD and calyx SASA already read the full 1000 ns for all replicas. This does not invalidate the existing results – values are internally consistent and match previously reported numbers – but the per-replica table below should be read with that caveat until the fix lands (see note under Table 1).

1. BLG Results

Headline numbers (unchanged, PBC-corrected, computed across all 4 replicas): 613 total contact events (97 in CENTER, 516 near-interface across replicas); 6 long contact events ($\geq$ 10 ns); calyx SASA range 24–37 nm² (mean 28.95 nm²); Pearson correlation between SASA and orientation angle $r = +0.006$ (block-bootstrap 95% CI [−0.09, +0.11]) – no coupling detected within these limits. These support the paper’s current framing: BLG’s calyx remains accessible throughout, but the protein does not commit to a bound, reoriented state within the simulated timescale.

Secondary structure content (DSSP) is stable across all 4 replicas: Helix 10.5–12.4%, Sheet 36.0–36.8%, Coil 50.8–52.9%. Hydrogen-bonding (CENTER only so far): protein–protein 107.3 ± 5.6 , protein–water 393.2 ± 13.2 , interface water–water 9539.6 ± 328.4 (mean counts per frame).

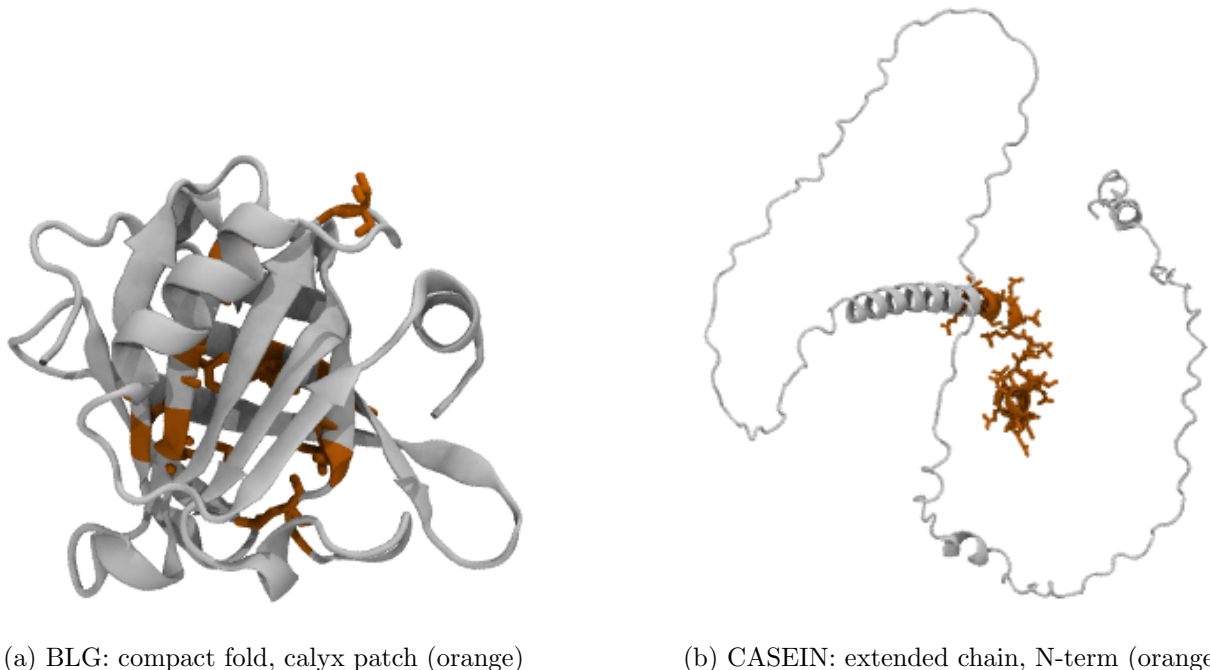


Figure 1: Structural contrast between the two proteins (VMD renders): BLG’s compact, mostly-buried calyx vs. CASEIN’s extended, disordered chain with its N-terminal region already surface-exposed even in the starting structure.

Replica	R_g (nm)*	RMSD bb (nm)	RMSD patch (nm)	Calyx SASA (nm ²)	γ (mN/m)*
CENTER (1000 ns)	1.504 ± 0.022	0.229	0.198	3.81 ± 0.46	51.9 ± 38.5
R1	1.497 ± 0.009	0.237	0.347	3.15 ± 0.34	52.7 ± 38.7
R2	1.493 ± 0.008	0.206	0.254	3.59 ± 0.32	52.0 ± 39.0
R3	1.499 ± 0.012	0.268	0.276	3.36 ± 0.48	52.6 ± 39.4

Table 1: BLG per-replica structural summary. *For R1/R2/R3, only R_g and γ (surface tension) currently reflect the first 500 of 1000 ns each replica actually ran – a fix to include the remaining trajectory segments for these two analyses is in progress. RMSD and calyx SASA for R1/R2/R3 already cover the full 1000 ns (their analysis scripts already include the extension trajectory segments).

2. CASEIN (β -casein) Results

First comparative result with real replication (n=2): N-terminal region (residues 1–25) solvent accessibility – the CASEIN analogue of BLG’s calyx SASA – is 24.77 ± 1.40 nm² in R1 vs. 25.41 ± 1.97 nm² in CENTER. The two independent 1000 ns runs agree closely, which is good evidence this is a real, reproducible property of the protein rather than an artifact of one starting structure.

CASEIN’s near-constant interfacial contact (99.5% of frames in *both* CENTER and R1, versus BLG’s much more selective, intermittent contact pattern across 4 μ s) is a clear structural contrast that directly supports the comparative title: BLG presents a small, selectively-accessible calyx, while CASEIN’s disordered chain stays persistently near the interface with its N-terminal region continuously exposed. R_g for R1 confirms the same relaxation pattern already seen in CENTER: an initial compaction from the extended AlphaFold starting structure (~ 2.85 nm in the first 100 ns) settling to a stable range (~ 2.5 nm) for the remainder of the trajectory – reassuring that this is a real, reproducible relaxation, not a one-off artifact of the CENTER run’s starting conformation. Hydrogen-bonding is also now complete for both replicas and agrees closely (protein-protein 92.0–97.8, protein-water 565.5–575.9, interface water-water 12015.6–12750.5 mean counts per frame) –

Replica	N-term SASA (nm ²)	Contact	γ (mN/m)	Hbonds prot-prot	Hbonds prot-water
CENTER (1000 ns)	25.41 $\pm$ 1.97	99.5%	51.4 $\pm$ 34.2	97.8 $\pm$ 9.7	565.5 $\pm$ 25.1
R1 (1000 ns)	24.77 $\pm$ 1.40	99.5%	52.4 $\pm$ 34.6	92.0 $\pm$ 8.3	575.9 $\pm$ 24.0

Table 2: CASEIN per-replica summary. Both replicas now have all 7 planned analyses complete; R1’s hydrogen-bonding analysis is the most recently finished. ”Contact” is the fraction of frames with a protein-water minimum distance ≤ 0.3 nm; R1 shows 1 sustained contact event across the full 1000 ns run.

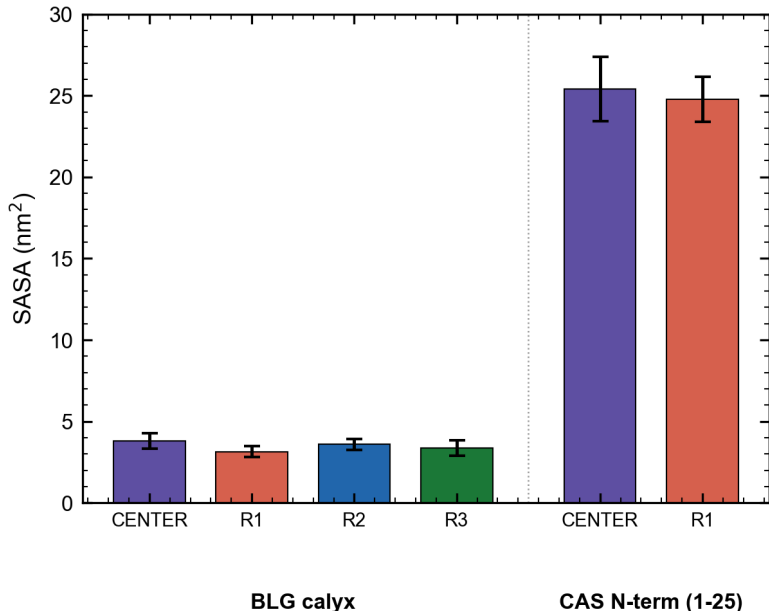


Figure 2: Accessible surface area, BLG calyx vs. CASEIN N-terminal region (residues 1–25), all replicas. The ~ 7 -fold size difference is consistent across every replica of both proteins – not an artifact of any single run.

another independent confirmation that CENTER’s behavior was not a one-off.

3. Questions for P.P.

1. **IFSC2026 (time-sensitive)** – the abstract deadline is Sept 25, 2026 – **14 days away as of this report**. We still have no confirmation whether participation is actually happening, or which framing (BLG-only complete story vs. BLG+CAS-in-progress) you’d prefer for the poster. This needs an answer this week to leave time to draft the abstract.
2. **”Secondary SASA” definition** – your June 9 note is ambiguous between per-secondary-structure-element SASA (computable now) and a Δ SASA-on-adsorption (not computable – no completed adsorption event exists in either dataset). We propose the former; a one-line confirmation from you would let us build it.
3. **Lab experiment correlation** – should the ”correlation with lab experiment” comparison be against published literature (already surveyed) or in-house wet-lab data? We see no evidence of in-house tensiometry capability in the group, so we’re assuming literature – please confirm.
4. **How prescriptive should the ”modify to adsorb” claim be?** We have data supporting a soft design-principle statement (disorder + exposed hydrophobic patches lower the kinetic barrier to adsorption), but not a strong causal claim, since neither dataset shows a completed adsorption event. Please confirm this is the right level of claim for Paper 1.

5. **Co-author review of the BLG-only draft** – still the hard blocker for eventual submission.
Whenever you have time for a pass, we're ready.

Note: all numbers above are PBC-corrected and read directly from current result files as of this report's generation date – treat as current, not frozen, if analysis is rerun later.